

to have affected about 8.3 million computer networks around the world, according to US research firm Computer Economics (November 2001). Nimda is a worm that uses multiple methods to automatically infect other computers. It can replicate through e-mail using an exploit that was made public months before Nimda hit, the MIME Header exploit. BadTrans.B is a mass-mailing worm that distributes itself using the MIME Header exploit. BadTrans.B first appeared after the Nimda outbreak. With their highly rapid infection rate, both Nimda and BadTrans.B took anti-virus vendors by surprise. Though the vendors tried to issue definition file updates as soon as they learned about each virus, the virus had already succeeded in infecting a large number of PCs by the time the anti-virus updates were released. Though both viruses used the same exploit, anti-virus vendors had to issue a separate definition file update for each. In contrast, an e-mail exploit detection engine would have recognized the exploit used and identified the attempt to automatically launch an executable file using the MIME header exploit. As a result, it would have blocked both worms automatically, preventing infection.

Other Examples Of Exploits

Double extension vulnerability viruses: Klez, Netsky and Lovegate.

What it does: Malicious files are given a double extension such as filename.txt.exe to trick the user into running the executable.

URL spoofing exploit viruses: No virus/worm has been found to be using this method. However it has been used to inject backdoors on Windows computers.

What it does: Allows spammers and phishers (scammers, or people trying to defraud computer users) to fool users to visit a malicious website instead of a legitimate one.

Object data file execution viruses: Bagle.Q.

What it does: Allows attackers to automatically infect unpatched versions of IE/Outlook (Express) by downloading and executing code from an HTTP site.